

ChatGPT

KL Divergence Explanation

[explain KL Divergence , its working and maths behind it.](#) 2



explain KL Divergence , its working and maths behind it

What is KL Divergence?

KL Divergence (Kullback–Leibler Divergence) measures **how different one probability distribution is from another**.

It answers:

“If the true distribution is P , how much information is lost when we approximate it using Q ?”

It is widely used in:

- Machine Learning
- Deep Learning
- Information Theory
- Variational Autoencoders (VAEs)
- Reinforcement Learning
- Language Models
- Bayesian inference

Intuition

Suppose:

- P = real/true distribution
- Q = predicted/approximate distribution

KL divergence tells us:

- how surprised we are when using Q instead of P
- how inefficient Q is in representing data generated from P

If $P = Q$:

$$D_{KL}(P||Q) = 0$$

The larger the difference, the larger the KL divergence.

Core Formula

For discrete probability distributions:

$$D_{KL}(P||Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$$

Where:

- $P(x)$ = true probability
- $Q(x)$ = approximate probability
- $\log$ is usually natural log

Understanding the Formula

Let's break this into parts.

1. Ratio Term

$$\frac{P(x)}{Q(x)}$$

This compares:

- what should happen
vs
- what our model believes

Case 1: Good prediction

If:

$$P(x) = Q(x)$$

then:

$$\log \frac{P(x)}{Q(x)} = \log(1) = 0$$

No penalty.

Case 2: Underestimating probability

If:

$$P(x) > Q(x)$$

then:

$$\log \frac{P(x)}{Q(x)} > 0$$

Large penalty.

The model assigned too little probability to something important.

Case 3: Overestimating probability

If:

$$Q(x) > P(x)$$

then log becomes negative.

But since everything is weighted by $P(x)$, the overall divergence still remains non-negative.

Why Multiply by $P(x)$?

$$P(x) \log \frac{P(x)}{Q(x)}$$

Because we care more about events that truly happen often.

Rare events contribute less.

This is like a weighted average of "surprise."

Connection to Information Theory

KL divergence comes from **entropy**.

Entropy

Entropy measures uncertainty:

$$H(P) = - \sum_x P(x) \log P(x)$$

High entropy:

- more randomness

Low entropy:

- more certainty
-

Cross Entropy

Cross entropy measures coding cost when using Q to encode data from P :

$$H(P, Q) = - \sum_x P(x) \log Q(x)$$

Relationship

KL divergence is:

$$D_{KL}(P||Q) = H(P, Q) - H(P)$$

Meaning:

Extra information/coding cost caused by using Q instead of the true distribution P .

Numerical Example

Suppose true distribution:

$$P = [0.8, 0.2]$$

Model prediction:

$$Q = [0.6, 0.4]$$

Then:

$$D_{KL}(P||Q) = 0.8 \log \frac{0.8}{0.6} + 0.2 \log \frac{0.2}{0.4}$$

Calculate:

$$= 0.8 \log(1.333) + 0.2 \log(0.5)$$

Using natural logs:

$$= 0.8(0.287) + 0.2(-0.693)$$

$$= 0.2296 - 0.1386$$

$$\approx 0.091$$

Small divergence $\rightarrow$ distributions are somewhat similar.

Important Properties

1. Non-negative

$$D_{KL}(P||Q) \geq 0$$

Always true.

Equal to 0 only when distributions are identical.

2. Not Symmetric

Very important:

$$D_{KL}(P||Q) \neq D_{KL}(Q||P)$$

So KL divergence is **not a true distance metric**.

Example intuition:

Missing an important event is worse than assigning extra probability somewhere else.

3. Can Become Infinite

If:

$$Q(x) = 0$$

while:

$$P(x) > 0$$

then:

$$\log \frac{P(x)}{0} \rightarrow \infty$$

Infinite divergence.

Meaning:

- the model says an event is impossible
- but it actually occurs

Huge error.

Geometric Interpretation

KL divergence measures:

how much one probability surface differs from another.

In optimization:

- minimizing KL divergence pushes predicted distributions toward target distributions.

Continuous Version

For continuous distributions:

$$D_{KL}(P||Q) = \int p(x) \log \frac{p(x)}{q(x)} dx$$

Used heavily in:

- Gaussian distributions
- Bayesian ML
- VAEs

KL Divergence in Machine Learning

1. Classification

Cross-entropy loss is directly related to KL divergence.

Training a classifier minimizes divergence between:

- true labels
- predicted probabilities

2. Variational Autoencoders (VAEs)

VAEs use KL divergence to force latent vectors toward a normal distribution.

Loss:

$$\text{Reconstruction Loss} + D_{KL}(q(z|x)||p(z))$$

This regularizes the latent space.

3. Reinforcement Learning

Used in:

- PPO
- Trust Region methods

to prevent policy updates from changing too drastically.

4. Language Models

Used when comparing:

- predicted token distributions
- target distributions

Forward vs Reverse KL

Forward KL

$$D_{KL}(P||Q)$$

Penalizes missing regions where P has probability mass.

Encourages covering all modes.

Reverse KL

$$D_{KL}(Q||P)$$

Penalizes assigning probability where P has none.

Encourages mode-seeking behavior.

Visual Intuition

Imagine two hills representing distributions.

KL divergence measures:

- how badly one hill approximates the other
- especially where the true hill is high

Relation to Other Metrics

Metric	Symmetric?	True Distance?
KL Divergence	No	No
Jensen-Shannon Divergence	Yes	Partially
Wasserstein Distance	Yes	Yes
Euclidean Distance	Yes	Yes

Why KL Divergence Matters

It provides a mathematically grounded way to compare uncertainty and probability distributions.

This is fundamental because modern AI systems:

- predict probabilities
- learn distributions
- optimize uncertainty

KL divergence is one of the core tools enabling that.

Self-information

Surprise of an outcome

$$I(x) = \log \frac{1}{P(x)}$$

Entropy

Average surprise

$$H(P) = \sum_i P(x_i) \log \frac{1}{P(x_i)}$$

Cross-Entropy

Average surprise under wrong belief

$$H(P, Q) = \sum_i P(x_i) \log \frac{1}{Q(x_i)}$$

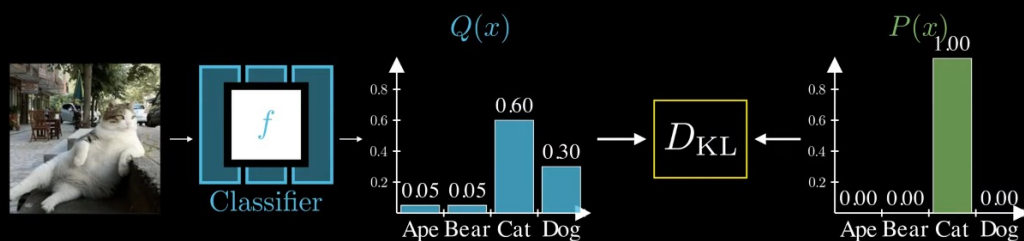
KL Divergence

Extra cost for wrong belief

$$D_{\text{KL}}(P \| Q) = \sum_i P(x_i) \log \frac{P(x_i)}{Q(x_i)}$$

$$D_{\text{KL}}(P \| Q) = \sum_i P(x_i) \log P(x_i) - \sum_i P(x_i) \log Q(x_i)$$

Constant



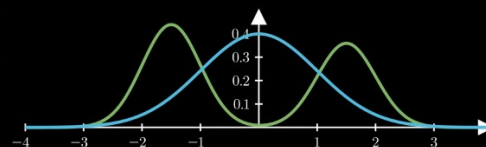
$P(x)$: True distribution (Reference)

$Q(x)$: Approximate distribution (Model)

Forward KL

Mode-covering

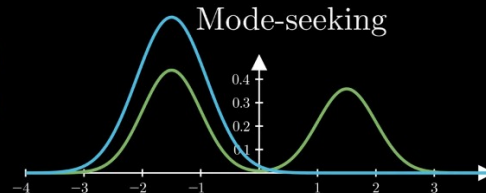
$$D_{\text{KL}}(P \| Q) = \sum_i P(x_i) \log \frac{P(x_i)}{Q(x_i)}$$



Reverse KL

Mode-seeking

$$D_{\text{KL}}(Q \| P) = \sum_i Q(x_i) \log \frac{Q(x_i)}{P(x_i)}$$





$$D_{\text{KL}}(P\|Q) = \int_{-\infty}^{\infty} P(x) \log \frac{P(x)}{Q(x)} dx$$

No closed-form solution

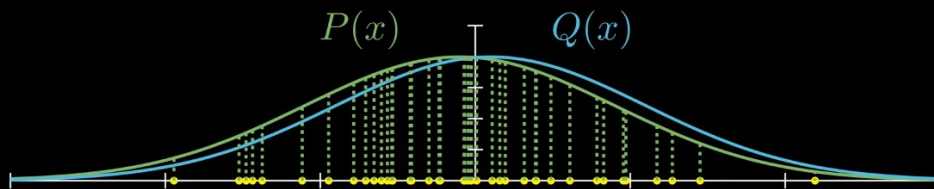
$$D_{\text{KL}}(P\|Q) = \mathbb{E}_{x \sim P} \left[\log \frac{P(x)}{Q(x)} \right]$$

Monte Carlo Estimation

Sample $x_1, x_2, \dots, x_N \sim P(x)$

True $D_{\text{KL}}(P\|Q) = 0.02$

$$\hat{D}_{\text{KL}}(P\|Q) = \frac{1}{N} \sum_{k=1}^N \log \frac{P(x_k)}{Q(x_k)}$$



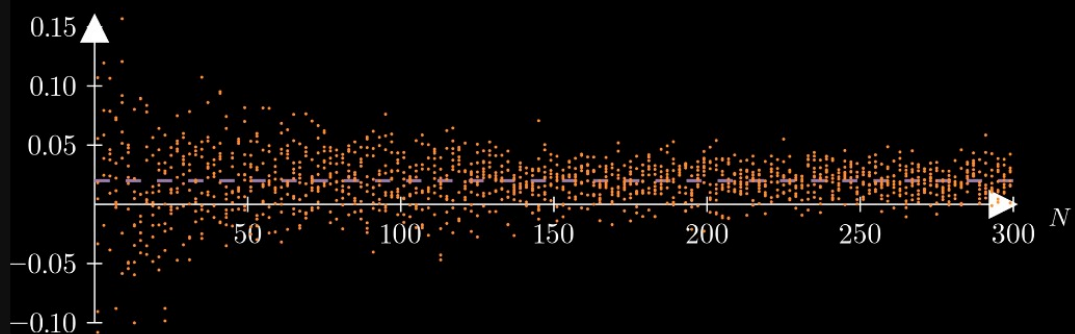
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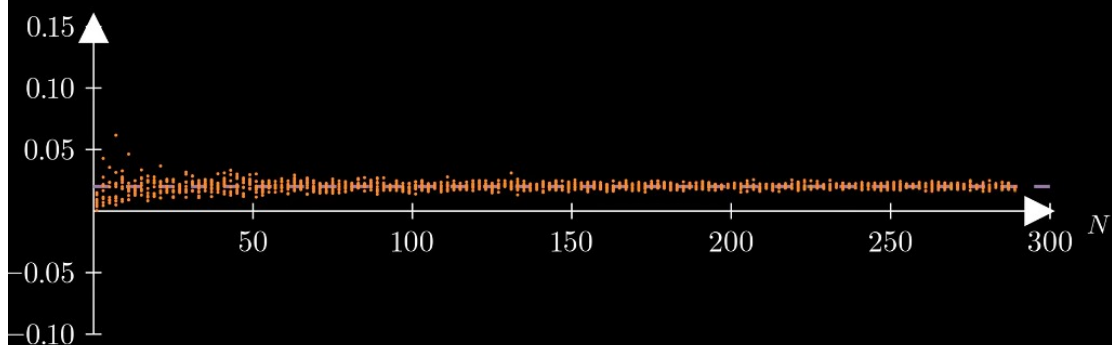
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Monte Carlo Estimation

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True $D_{\text{KL}}(P\|Q) = 0.02$

$$\hat{D}_{\text{KL}}(P\|Q) = \frac{1}{N} \sum_{k=1}^N \frac{1}{2} \left(\log \frac{P(x_k)}{Q(x_k)} \right)^2$$



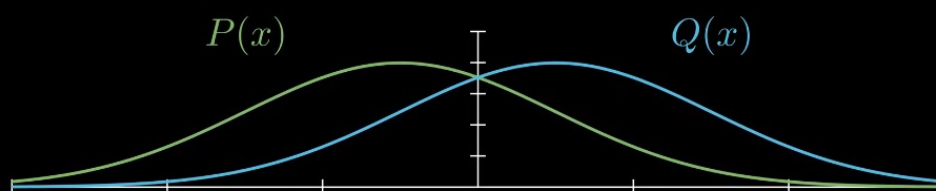
$$D_{\text{KL}}(P\|Q) = \mathbb{E}_{x \sim P} \left[\log \frac{P(x)}{Q(x)} \right]$$

Monte Carlo Estimation

Sample $x_1, x_2, \dots, x_N \sim P(x)$

True $D_{\text{KL}}(P\|Q) = 0.5$

$$\hat{D}_{\text{KL}}(P\|Q) = \frac{1}{N} \sum_{k=1}^N \frac{1}{2} \left(\log \frac{P(x_k)}{Q(x_k)} \right)^2$$



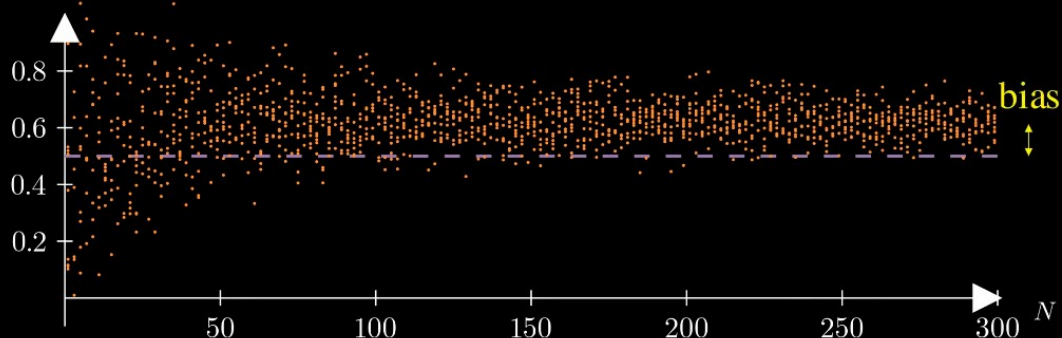
$$D_{\text{KL}}(P\|Q) = \mathbb{E}_{x \sim P} \left[\log \frac{P(x)}{Q(x)} \right]$$

Monte Carlo Estimation

Sample $x_1, x_2, \dots, x_N \sim P(x)$

True $D_{\text{KL}}(P\|Q) = 0.5$

$$\hat{D}_{\text{KL}}(P\|Q) = \frac{1}{N} \sum_{k=1}^N \frac{1}{2} \left(\log \frac{P(x_k)}{Q(x_k)} \right)^2$$



$$D_{\text{KL}}(P\|Q) = \mathbb{E}_{x \sim P} \left[\log \frac{P(x)}{Q(x)} \right]$$

Monte Carlo Estimation

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$$\hat{D}_{\text{KL}}(P\|Q) = \frac{1}{N} \sum_{k=1}^N \frac{1}{2} \left(\log \frac{P(x_k)}{Q(x_k)} \right)^2$$



Unbiased



Biased



High variance



Low variance

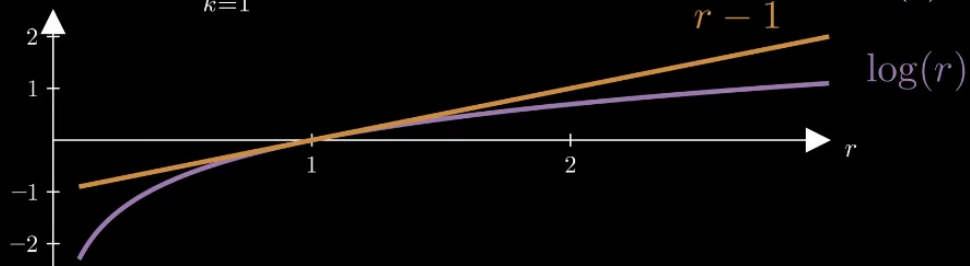
$$D_{\text{KL}}(P\|Q) = \mathbb{E}_{x \sim P} \left[\log \frac{P(x)}{Q(x)} \right]$$

Monte Carlo Estimation

Sample $x_1, x_2, \dots, x_N \sim P(x)$

$$\hat{D}_{\text{KL}}(P\|Q) = \frac{1}{N} \sum_{k=1}^N [r(x_k) - 1 - \log r(x_k)]$$

$$r(x) = \frac{Q(x)}{P(x)}$$



$$D_{\text{KL}}(P\|Q) = \mathbb{E}_{x \sim P} \left[\log \frac{P(x)}{Q(x)} \right]$$

$$r(x) = \frac{Q(x)}{P(x)}$$

Monte Carlo Estimation

Sample $x_1, x_2, \dots, x_N \sim P(x)$

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